ANSWER KEY & MARKING SCHEME — SET 1

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 7 m	$h = 7\tan 45^\circ = 7 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = 7 m
2	**(b)** $5\sqrt{3}$ m	$h = 5\tan 60^\circ = 5 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = $5\sqrt{3}$ m
3	**(c)** $3\sqrt{3}$ m	$h = 9\tan 30^\circ = 9 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = $3\sqrt{3}$ m
4	**(d)** 9 m	$x = 9\cot 45^\circ$; Shadow = 9 m
5	**(a)** 12 m	$\tan 45^\circ = \frac{12}{x}$ so $x = 12\cot 45^\circ$; Distance = 12 m
6	**(c)** $4\sqrt{3}$ m	Height = $8\sin 60^\circ = 8 \times \frac{\sqrt{3}}{2}$; Height reached on the wall = $4\sqrt{3}$ m
7	**(c)** line of sight	Definition
8	**(a)** 12 m	$h = 12\tan 45^\circ = 12 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = 12 m
9	**(d)** $8\sqrt{3}$ m	$h = 8\tan 60^\circ = 8 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = $8\sqrt{3}$ m
10	**(a)** $4\sqrt{3}$ m	$h = 12\tan 30^\circ = 12 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = $4\sqrt{3}$ m

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let the height be h; in the right triangle, $\tan 45^\circ = \frac{h}{10}$; $h = 10\tan 45^\circ = 10 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$) (1); Height = 10 m (1).	1+1
12	Let the shadow be x; $\tan 45^\circ = \frac{14}{x}$; $x = 14\cot 45^\circ$ (1); Shadow = 14 m (1).	1+1
13	Kite string is the hypotenuse of the right triangle; $h = 20\sin 30^\circ = 20 \times \frac{1}{2}$ (1); Height = 10 m (1).	1+1
14	Let the height be h; in the right triangle, $\tan 45^\circ = \frac{h}{16}$; $h = 16\tan 45^\circ = 16 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$) (1); Height = 16 m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ladder is the hypotenuse; height = $12\sin 60^\circ$, base = $12\cos 60^\circ$ (1); $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$ (1); Height = $6\sqrt{3}$ m, distance = 6 m (1).	1+1+1
16	Angle of depression = angle of elevation of the top of the cliff from the boat = 60° (1); $\tan 60^\circ = \frac{10\sqrt{3}}{x}$ so $x = 10\sqrt{3}\cot 60^\circ$ (1); Distance = 10 m (1).	1+1+1
17	Let the shadow be x; $\tan 60^\circ = \frac{12\sqrt{3}}{x}$ (1); $x = 12\sqrt{3}\cot 60^\circ$ (1); Shadow = 12 m (1).	1+1+1
	OR: Ladder is the hypotenuse; height = $16\sin 60^\circ$, base = $16\cos 60^\circ$ (1); $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$ (1); Height = $8\sqrt{3}$ m, distance = 8 m (1).	1+1+1
18	Angle of depression = angle of elevation of the top of the cliff from the boat = 60° (1); $\tan 60^\circ = \frac{15\sqrt{3}}{x}$ so $x = 15\sqrt{3}\cot 60^\circ$ (1); Distance = 15 m (1).	1+1+1
	OR: Let the shadow be x; $\tan 60^\circ = \frac{9\sqrt{3}}{x}$ (1); $x = 9\sqrt{3}\cot 60^\circ$ (1); Shadow = 9 m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Distances of the points from the foot: $h\cot 30^\circ = 18$ m and $h\cot 60^\circ = 6$ m (2); The points are on opposite sides, so the distances add (2); Distance = 24 m (1).	2+2+1
	OR: Let height = h; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 6 m (2); $h(\cot 30^\circ - \cot 60^\circ) = 6$ gives $h = 6 \div (\frac{2\sqrt{3}}{3})$ (2); Height = $3\sqrt{3}$ m (distances 9 m and 3 m) (1).	2+2+1
20	Distances of the points from the foot: $h\cot 30^\circ = 27$ m and $h\cot 60^\circ = 9$ m (2); The points are on opposite sides, so the distances add (2); Distance = 36 m (1).	2+2+1
	OR: Let height = h; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 10 m (2); $h(\cot 30^\circ - \cot 60^\circ) = 10$ gives $h = 10 \div (\frac{2\sqrt{3}}{3})$ (2); Height = $5\sqrt{3}$ m (distances 15 m and 5 m) (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 15 m	$x = 15\cot 45^\circ$; Shadow $= 15$ m
2	**(c)** 20 m	$\tan 45^\circ = \frac{20}{x}$ so $x = 20\cot 45^\circ$; Distance $= 20$ m
3	**(c)** $5\sqrt{3}$ m	Height $= 10\sin 60^\circ = 10 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 5\sqrt{3}$ m
4	**(c)** the horizontal line upwards to the line of sight	Definition
5	**(d)** 15 m	$h = 15\tan 45^\circ = 15 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 15$ m
6	**(b)** $11\sqrt{3}$ m	$h = 11\tan 60^\circ = 11 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 11\sqrt{3}$ m
7	**(b)** $6\sqrt{3}$ m	$h = 18\tan 30^\circ = 18 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 6\sqrt{3}$ m
8	**(d)** $8\sqrt{3}$ m	$x = 8\cot 30^\circ$; Shadow $= 8\sqrt{3}$ m
9	**(c)** $7\sqrt{3}$ m	$\tan 30^\circ = \frac{7}{x}$ so $x = 7\cot 30^\circ$; Distance $= 7\sqrt{3}$ m
10	**(c)** 7 m	Height $= 14\sin 30^\circ = 14 \times \frac{1}{2}$; Height reached on the wall $= 7$ m

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let the shadow be x; $\tan 30^\circ = \frac{11}{x}$; $x = 11\cot 30^\circ$ (1); Shadow $= 11\sqrt{3}$ m (1).	1+1
12	Kite string is the hypotenuse of the right triangle; $h = 36\sin 30^\circ = 36 \times \frac{1}{2}$ (1); Height $= 18$ m (1).	1+1
13	Let the height be h; in the right triangle, $\tan 60^\circ = \frac{h}{4}$; $h = 4\tan 60^\circ = 4 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$) (1); Height $= 4\sqrt{3}$ m (1).	1+1
14	Let the shadow be x; $\tan 45^\circ = \frac{7}{x}$; $x = 7\cot 45^\circ$ (1); Shadow $= 7$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ladder is the hypotenuse; height $= 22\sin 30^\circ$, base $= 22\cos 30^\circ$ (1); $\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$ (1); Height $= 11$ m, distance $= 11\sqrt{3}$ m (1).	1+1+1
16	Angle of depression $=$ angle of elevation of the top of the cliff from the boat $= 30^\circ$ (1); $\tan 30^\circ = \frac{4\sqrt{3}}{x}$ so $x = 4\sqrt{3}\cot 30^\circ$ (1); Distance $= 12$ m (1).	1+1+1
17	Let the shadow be x; $\tan 30^\circ = \frac{5\sqrt{3}}{x}$ (1); $x = 5\sqrt{3}\cot 30^\circ$ (1); Shadow $= 15$ m (1).	1+1+1
	OR: Ladder is the hypotenuse; height $= 26\sin 30^\circ$, base $= 26\cos 30^\circ$ (1); $\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$ (1); Height $= 13$ m, distance $= 13\sqrt{3}$ m (1).	1+1+1
18	Angle of depression $=$ angle of elevation of the top of the cliff from the boat $= 60^\circ$ (1); $\tan 60^\circ = \frac{6\sqrt{3}}{x}$ so $x = 6\sqrt{3}\cot 60^\circ$ (1); Distance $= 6$ m (1).	1+1+1
	OR: Let the shadow be x; $\tan 60^\circ = \frac{14\sqrt{3}}{x}$ (1); $x = 14\sqrt{3}\cot 60^\circ$ (1); Shadow $= 14$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Distances of the points from the foot: $h\cot 30^\circ = 36$ m and $h\cot 60^\circ = 12$ m (2); The points are on opposite sides, so the distances add (2); Distance $= 48$ m (1).	2+2+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 14 m (2); $h(\cot 30^\circ - \cot 60^\circ) = 14$ gives $h = 14 \div (\frac{2\sqrt{3}}{3})$ (2); Height $= 7\sqrt{3}$ m (distances 21 m and 7 m) (1).	2+2+1
20	Distances of the points from the foot: $h\cot 30^\circ = 45$ m and $h\cot 60^\circ = 15$ m (2); The points are on opposite sides, so the distances add (2); Distance $= 60$ m (1).	2+2+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 16 m (2); $h(\cot 30^\circ - \cot 60^\circ) = 16$ gives $h = 16 \div (\frac{2\sqrt{3}}{3})$ (2); Height $= 8\sqrt{3}$ m (distances 24 m and 8 m) (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 30°	Alternate angles are equal
2	**(d)** 20 m	$h = 20\tan 45^\circ = 20 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 20$ m
3	**(a)** $14\sqrt{3}$ m	$h = 14\tan 60^\circ = 14 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 14\sqrt{3}$ m
4	**(c)** $8\sqrt{3}$ m	$h = 24\tan 30^\circ = 24 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 8\sqrt{3}$ m
5	**(b)** $10\sqrt{3}$ m	$x = 10\cot 30^\circ$; Shadow $= 10\sqrt{3}$ m
6	**(b)** $15\sqrt{3}$ m	$\tan 30^\circ = \frac{15}{x}$ so $x = 15\cot 30^\circ$; Distance $= 15\sqrt{3}$ m
7	**(a)** 9 m	Height $= 18\sin 30^\circ = 18 \times \frac{1}{2}$; Height reached on the wall $= 9$ m
8	**(b)** becomes shorter	Larger elevation gives smaller $\cot$
9	**(c)** 25 m	$h = 25\tan 45^\circ = 25 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 25$ m
10	**(d)** $25\sqrt{3}$ m	$h = 25\tan 60^\circ = 25 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 25\sqrt{3}$ m

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Kite string is the hypotenuse of the right triangle; $h = 24\sin 60^\circ = 24 \times \frac{\sqrt{3}}{2}$ (1); Height $= 12\sqrt{3}$ m (1).	1+1
12	Let the height be h; in the right triangle, $\tan 60^\circ = \frac{h}{13}$; $h = 13\tan 60^\circ = 13 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$) (1); Height $= 13\sqrt{3}$ m (1).	1+1
13	Let the shadow be x; $\tan 30^\circ = \frac{12}{x}$; $x = 12\cot 30^\circ$ (1); Shadow $= 12\sqrt{3}$ m (1).	1+1
14	Kite string is the hypotenuse of the right triangle; $h = 30\sin 60^\circ = 30 \times \frac{\sqrt{3}}{2}$ (1); Height $= 15\sqrt{3}$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ladder is the hypotenuse; height $= 30\sin 60^\circ$, base $= 30\cos 60^\circ$ (1); $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$ (1); Height $= 15\sqrt{3}$ m, distance $= 15$ m (1).	1+1+1
16	Angle of depression $=$ angle of elevation of the top of the cliff from the boat $= 30^\circ$ (1); $\tan 30^\circ = \frac{20\sqrt{3}}{x}$ so $x = 20\sqrt{3}\cot 30^\circ$ (1); Distance $= 60$ m (1).	1+1+1
17	Let the shadow be x; $\tan 30^\circ = \frac{8\sqrt{3}}{x}$ (1); $x = 8\sqrt{3}\cot 30^\circ$ (1); Shadow $= 24$ m (1).	1+1+1
	OR: Ladder is the hypotenuse; height $= 40\sin 30^\circ$, base $= 40\cos 30^\circ$ (1); $\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$ (1); Height $= 20$ m, distance $= 20\sqrt{3}$ m (1).	1+1+1
18	Angle of depression $=$ angle of elevation of the top of the cliff from the boat $= 30^\circ$ (1); $\tan 30^\circ = \frac{9\sqrt{3}}{x}$ so $x = 9\sqrt{3}\cot 30^\circ$ (1); Distance $= 27$ m (1).	1+1+1
	OR: Let the shadow be x; $\tan 60^\circ = \frac{11\sqrt{3}}{x}$ (1); $x = 11\sqrt{3}\cot 60^\circ$ (1); Shadow $= 11$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Distances of the points from the foot: $h\cot 30^\circ = 54$ m and $h\cot 60^\circ = 18$ m (2); The points are on opposite sides, so the distances add (2); Distance $= 72$ m (1).	2+2+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 18 m (2); $h(\cot 30^\circ - \cot 60^\circ) = 18$ gives $h = 18 \div (\frac{2\sqrt{3}}{3})$ (2); Height $= 9\sqrt{3}$ m (distances 27 m and 9 m) (1).	2+2+1
20	Distances of the points from the foot: $h\cot 30^\circ = 63$ m and $h\cot 60^\circ = 21$ m (2); The points are on opposite sides, so the distances add (2); Distance $= 84$ m (1).	2+2+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 22 m (2); $h(\cot 30^\circ - \cot 60^\circ) = 22$ gives $h = 22 \div (\frac{2\sqrt{3}}{3})$ (2); Height $= 11\sqrt{3}$ m (distances 33 m and 11 m) (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

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